Blockchain-based Life Cycle Assessment System for ESG Reporting

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Abstract

With the increasing popularity of Environmental, Social, and Governance (ESG) reporting, doubts about the reliability and completeness of ESG disclosures also grow. Those concerns may be alleviated by Life Cycle Assessment (LCA), a widely used approach for a complete and comprehensive evaluation of environmental impacts. However, LCA is limited in the areas of reliable data collection and management. Blockchain can solve this problem by reliably storing and distributing data from various sources without tampering via its decentralized and secure nature. This study aims to explore the influence of LCA and blockchain on ESG reporting. We propose a blockchain-based LCA system and demonstrate its potential applications using data regarding Tesla's electric vehicles. The system automatically cross-validates ESG disclosures from companies of an entire value chain and generates an integrated report to present overall ESG performance. Our results shed light on utilizing emerging technologies to enable reliable, complete, and accurate ESG reporting.

Keywords: ESG, Life Cycle Assessment (LCA), Blockchain, Smart contract, IoT

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1 Introduction

The last few decades have seen a growing interest in Environmental, Social, and Governance (ESG) information from a variety of stakeholders such as firms, regulators, and investors (Inderst and Stewart 2018; Amel-Zadeh and Serafeim 2018; Miralles-Quirós et al. 2019; Gillan et al. 2021). However, the reliability of ESG disclosures is commonly in doubt due to companies' discretion in choosing what to report, the lack of transparency in ESG measuring and reporting approaches, and the lack of adoption and low quality of ESG assurance (Talbot and Boiral 2018; Boiral and Heras-Saizarbitoria 2020). GHG reporting and impression management: An assessment of sustainability reports from the energy sector. Journal of Business Ethics, 147(2), 367-383.). A fraudulent issue related to environmental disclosure, namely greenwashing[1], recently has become a major concern of ESG reports. Greenwashing is the selective disclosure of positive information about a company's ESG performance without full disclosure of negative information to create an overly positive corporate image (Lyon and Maxwell 2011). Recently, Tesla's ESG score fell to the bottom 25% of its sector group, and Tesla was removed from the S&P 500 ESG Index due to its lack of a low-carbon strategy.[2]

Consider this: an automobile manufacturer claims its new cars emit only 3.2 metric tons of carbon dioxide per car/year in their use stage. This does not necessarily mean that those cars have become more environmentally friendly than prior models since the company does not include carbon dioxide emissions at the manufacturing and disposal stages. It's possible that more energy or materials are required to manufacture the new cars, which are acquired at the cost of more carbon dioxide, or that the disposal of the new cars generates more carbon dioxide. A recent report issued by LowCVP documented that some of the CO2 savings made during the use of electric vehicles in comparison with standard vehicles were offset by increased emissions created during their production and, to a lesser extent, during disposal.[3] In addition, new components such as batteries of electric cars may result in severe pollution if not appropriately disposed of. It is estimated that more than 12 million tons of lithium-ion batteries are expected to retire between now and 2030, threatening to leave a mountain of electronic waste when they reach the end of their lives[4]. However, companies may greenwash their ESG reports by emphasizing only emission reduction when using cars while ignoring the emissions and potential pollution at the manufacturing and disposal stages.

Life Cycle Assessment (LCA) offers a solution to this problem by examining the environmental impacts of a product by accounting for all processes from inputs to outputs throughout the product's life cycle. Using ESG-related data from individual companies within a value chain, LCA can integrate them and assess the ESG performance of the entire chain. However, reliable and effective data gathering across the value chain has been identified as a big challenge for performing LCA (Zhang et al. 2020). For example, data fed into LCA models may be out of date. Certain data are difficult to collect, so they can only be estimated, but the estimation methods are misleading or not well described. Site-specific data are needed, but only average data from different sites are often available. Without high-quality data and transparent input processes, LCA may produce biased, incomplete, or fraudulent results and further significantly affect the credibility of LCA outcomes (Schryver et al. 2013). Blockchain and the Internet of Things (IoT) technologies can greatly alleviate this problem. IoT devices can collect ESG-related data from various sites in a value chain in real time, thus solving the data availability issue. Then, leveraging blockchain's characteristics of decentralization, transparency, trustworthiness, and immutability, the ESG-related data captured from IoT can be uploaded to blockchain, which can protect the security of the data and prevent management manipulation. Embedding smart contracts into blockchain can further facilitate cross-validation of ESG disclosures from each company and generate an integrated report to present the ESG performance of the entire value chain.

This paper aims to explore the potential use of LCA and blockchain on ESG reporting. Built on the prior literature, we propose a blockchain-based LCA system to improve the reliability, completeness, and accuracy of ESG reports. The system can generate an integrated report that presents the overall ESG performance of an entire value chain. It can also cross-validate ESG information reported by individual companies within the chain as well as validate a company's ESG report with its LCA outcome. Then, the proposed system is demonstrated using the real data of Tesla's electric vehicles collected in their life cycle. In the demonstration, a smart contract is created on Ethereum, a popular blockchain platform, to validate data inputs for LCA. Next, the validated ESG-related data are posted on the Ethereum blockchain, which are securely shared among stakeholders and used as inputs for a popular LCA software. Then, the LCA software assesses the environmental impact of Tesla's electric vehicles in each stage of their life cycle. Finally, a potential integrated ESG report that reflects the overall environmental impact of Tesla's electric vehicles is generated based on the LCA results.

The contributions of this paper are fourfold. First, this paper proposes a system to achieve a synergy effect of two technologies, blockchain and LCA, on facilitating ESG reporting with novel applications, i.e., cross-validation of ESG information from different companies, validation of a company's ESG report with its LCA outcome, and integrated ESG reports for an entire value chain. The proposed system will be of interest to stakeholders (e.g., regulators, NGOs, consumers, and investors) who are concerned about companies' activities on global climate change and other environment issues. It offers a reliable summary of emissions from an entire value chain and in a timely manner traces major emission sources, enabling more proactive actions to be taken to reduce the climate change risk. It also aids auditors in assurance tests by providing original and non-tampered data. Second, the proposed system and demonstration using Tesla's electric vehicles provide real-life insights and lessons on the incorporation of LCA and blockchain into ESG reporting. Third, this paper discusses the challenges in the system adoption, as well as potential solutions that could mitigate the concerns. Finally, this paper echoes the Center for Audit Quality's call for researching nonfinancial information disclosure (CAQ, 2021).

The rest of the paper is structured as follows: Section 2 reviews the basic concepts and literature of ESG, LCA, blockchain, and smart contracts. Section 3 describes the design of a system that incorporates LCA and blockchain to facilitate ESG reporting. Section 4 illustrates the implementation of the proposed system using real-life data of Tesla's electric vehicles. Section 5 discusses the strengths and potential hindrances to the full acceptance of the system and identifies future research questions. The last section concludes the paper.

2 Background and Related Research

2.1 ESG

Over the last few decades, there has been a growing interest in ESG information from firms, business presses, regulators, and investors (Inderst and Stewart 2018;

Amel-Zadeh and Serafeim 2018; Miralles-Quirós et al. 2019; Gillan et al. 2021). According to IFAC, AICPA, and CIMA (2021), 91% of companies across twenty-two jurisdictions published sustainability information.

In most countries, companies have a lot of discretion in choosing what to disclose in ESG reports since they are not mandated. As a result, management could choose one or more from a variety of ESG reporting standards to follow. According to the European Financial Reporting Advisory Group (2021)[5], while the Global Reporting Initiative (GRI) Standard is most widely used (54% of companies use it), many companies use more than three standards. The discretion of ESG reporting leaves room for greenwashing, an increasing practice that companies utilize to build a good, environmentally-friendly image by disclosing misleading ESG information. Talbot and Boiral (2018) identify a variety of greenwashing methods and show that it is rather difficult for stakeholders to reasonably assess, monitor, and compare companies' ESG performance based on their ESG reports. In addition to the discretion of ESG reporting, the lack of transparency in companies' ESG reporting further leads to doubts about the reliability of ESG information.

Although assurance could help detect greenwashing and verify ESG reports, only 51% of ESG information is assured (IFAC, AICPA, and CIMA, 2021). Meanwhile, the quality of ESG assurance is also in doubt due to narrowed service and untransparent assurance processes (Gürtürk and Hahn 2016). Moreover, assurance statements are often boilerplate and do not generate from a material, substantial, and credible verification process (Boiral and Heras-Saizarbitoria 2020).

2.2 LCA

LCA is a methodology that estimates cumulative environmental impacts associated with manufacturing a product or delivering a service, thereby providing a comprehensive view of the potential tradeoff in environmental impacts from a given activity (Gonzales 2018; Linkov et al. 2017). LCA examines the impacts of a product on the environment by accounting for all processes from inputs to outputs throughout the product's life cycle. Stages of the product life cycle include its birth, design, raw material extraction, material production, use, and final disposal (Gonzales 2018). Prior literature has thoroughly explored the use of LCA in assessing products' environmental impacts (Corcelli et al. 2019; Krishna et al. 2017; UNEP 2012). For example, Wong et al. (2021) review carbon footprint assessment and LCA procedures for two types of vehicle products: Electric Vehicles (EVs) and Hydrogen Fuel Cell Vehicles (HFCVs). They compare the carbon footprint of a Tesla Model 3 EV and a Toyota MIRAI HFCV using the LCA software GREET. Pintilie et al. (2016) use LCA to identify and quantify the main environmental contributors from urban waste water and water reclamation in Spain. Pierobon et al. (2019) use LCA software to evaluate the embodied emissions and energy associated with building materials, manufacturing, and construction. Yin et al. (2021) draw upon data disclosed by vehicle manufacturers and use the GREET software to compare the carbon footprint of vehicles with different fuel types, including traditional fuel vehicles, EVs, and HFCVs. Their results show that the fuel production and consumption phases significantly contribute to the carbon footprint of all vehicles. They also find that although vehicle manufacturers often claim that the scope of assessment includes all energy, materials, substances, and processes, their reports present only specific cases but not detailed data that can be used for comparison across different manufacturers. They conclude that vehicle manufacturers need to disclose relevant data on carbon footprints more transparently to allow comparisons of their vehicles' emissions.

While LCA demonstrates its usefulness in assessing environmental impacts, prior studies also have discovered some limitations in its applications (Ekener et al. 2018). On the one hand, gathering all necessary data to be used in an assessment from their original sources is a challenging task and can be prohibitively expensive (Fauzi et al. 2019). Approximately 70–80% of the time and costs of performing an LCA are attributed to data gathering (Miah et al. 2018), especially with full LCA undertakings (Yavuz et al. 2018). Costs of LCA significantly increase when a large amount of data are required and from a variety of sources (van der Meer 2018), an issue also evidenced in the case study in our paper. If data are not available, they will either be estimated or averages will be used (Gonzales 2018). However, those approaches could oversimplify real-world problems and raise doubts about LCA results (van der Meer 2018). On the other hand, even though companies claim that their disclosure regarding environmental impacts of products is based on sustainability standards, verification of those claims requires a costly and time-consuming auditing process (Mieras et al. 2019). Such issues will significantly impact the credibility and usefulness of LCA results in decision-making.

2.3 Blockchain

Blockchain was first proposed by Nakamoto (2008) as a secure infrastructure for cryptocurrency trading that can reduce trading costs and fraud risk as well as increase the effectiveness of auditing and monitoring (Swan 2015a; Fanning and Centers 2016; Pilkington 2016; Yermack 2017). In the past decade, blockchain has evolved from a cryptocurrency transaction system into a decentralized and distributed computing infrastructure that aims to verify and store data with an encrypted block structure, generate and update data with a distributed consensus algorithm, and process and control data with computer codes (e.g., smart contracts) (Qin et al. 2018; Qin et al.

2019). It has been combined with Artificial Intelligence (AI), the Internet of Things (IoT), and robotics to form an ecosystem of future commerce (Omohundro 2014; Dorri et al. 2016; Ferrer 2016).

2.3.1 Characteristics of Blockchain

Prior literature (Drescher 2017; Hughes et al. 2019; Yuan and Wang 2016; Sharif and Ghodoosi 2022) has identified the following key characteristics of blockchain.

1) Decentralized: Blockchain is a decentralized and trustworthy system that uses a mathematical method instead of a central organization to establish trust among distributed nodes.
2) Append-only and immutable: Data can be added to blockchain only in a sequential order. Once a transaction has been added to the blockchain, it cannot be altered.
3) Traceable: Blockchain is basically a "chain of blocks" in which each block stores the details and the timestamps of transactions. As a result, it offers a comprehensive audit trail that records the milestones an asset has crossed through a value chain, thus adding to the verifiability and traceability of data.
4) Collectively maintained: An economic incentive mechanism ensures that all nodes in a blockchain system participate in the verification processes of transactions and add new blocks to the blockchain through a consensus algorithm.
5) Open and transparent: All nodes in the network share the same master records.
6) Programmable: Blockchain may provide a flexible scripting system to create customized applications. For example, the Ethereum[6] blockchain platform provides a Turing complete script language, Solidity[7], for users to build smart

contracts.

7) Non-repudiate: Blockchain encrypts the data with public-key cryptography to reduce the risk of illicit transactions.

Due to these characteristics, blockchain could ensure the accuracy and reliability of ESG-related records by simplifying the processes of confirmation and comparing data from related entities, and it will enable real-time disclosure. Thus, blockchain could make the ESG reporting more complete, transparent, and precise.

2.3.2 Smart Contracts on Blockchain

Smart contracts are user-defined computer programs that specify rules agreed upon among entities and that operate on a blockchain (Delmolino et al. 2015; Kiviat 2015; Peters and Panayi 2016; Zhang et al. 2016). Upon deployment on a blockchain, a smart contract is distributed and collectively executed by all nodes in the blockchain. The majority of nodes then reach a consensus, eliminating fraudulent results from one or a few malicious nodes. The programmability of smart contracts greatly increases the flexibility and applications of blockchain (Dai and Vasarhelyi 2017), enhances the accuracy of contract execution, and enables enterprises to achieve real-time management and decision-making (Qin et al. 2020).

Existing accounting literature focuses on whether blockchain and smart contracts are suitable for integrating or replacing certain audit procedures (Dai et al. 2019). Dai and Vasarhelyi (2017) illustrate the potential roles of blockchain and smart contracts in both internal and external audits and discuss the challenges of involving them in the audit practice. Rozario and Vasarhelyi (2018) propose to automate audit data analytics by embedding certain analytical procedures into smart contracts. However, there is a lack of discussion on using blockchain-based smart contracts in ESG reporting and assurance.

2.4 The Impact of LCA and Blockchain on ESG Reporting

As discussed earlier, the reliability and accuracy of LCA results highly depend on the quality of input data (von Bahr and Steen 2004), and collecting reliable data has been a significant challenge in LCA (Hospido et al. 2010). Incorporating blockchain with LCA may help alleviate the problem since it can drastically improve the effectiveness and efficiency of collecting, analyzing, and disseminating necessary data to perform LCA. However, little research has explored the use of blockchain for LCA, with only a few exceptions (Krishna and Manickam 2017; Zhang et al. 2020; Teh et al. 2020; Dai et al. 2019).

Krishna and Manickam (2017) integrate blockchain and other advanced technologies into a standard LCA process to enhance its efficiency and effectiveness. Blockchain-based LCA ensures that all the important data elements are properly assessed and only validated results are used to draw conclusions (Rebitzer et al. 2004; Buyle et al. 2013; Krishna and Manickam 2017). Teh et al. (2020) explore how blockchain can address challenges that companies in the materials industry face in implementing LCA and facilitate a more effective LCA as part of their eco-sustainability policy and strategy. Zhang et al. (2020) design a blockchain-based LCA system for managing the environmental performance of companies within a supply chain. However, their frameworks do not deal with problems of storage scalability and possible data manipulation/error before being uploaded to blockchain.. We try to solve the problems by designing a blockchain-based life cycle assessment system, which directly uploads data collected from sensors to the blockchain and utilizes not only blockchain but also off-chain databases such as Enterprise Resource Planning (ERP) systems.

3 Blockchain-based Life Cycle Assessment System

Our system is built on the aforementioned blockchain-based LCA Framework (Krishna and Manickam 2017) and the framework proposed by Zhang et al. (2020). However, our system makes new contributions by focusing on the purpose of ESG reporting and proposing new applications, including cross-validation of ESG information among companies, integrated reporting of the entire value chain, and validation of a company's ESG report with its LCA outcome.

Figure 1 shows the proposed system and its applications. The life cycle component at the bottom articulates each stage in a product life cycle, i.e., raw material extraction, manufacturing, delivery, operation, and end-of-life (recycling or disposal). It serves as a guide for the system regarding where data collection, analysis, and assessment of a product's impacts on the environment should be performed. Taking an electric vehicle as an example, at the beginning of its life cycle, the extraction of raw materials such as steel and lithium consumes energy and produces emissions. Next, materials and electricity are consumed to manufacture the vehicle. The vehicle is then transported to customers and used, which also consumes electricity. Later, after being driven for a certain amount of miles, its batteries, tires, and parts are either disposed of or recycled, the former generating emissions and waste, while the latter also consumes electricity or other resources and generates emissions. Based on each stage of the product life cycle, the blockchain-based LCA system collects material and energy data and estimates emissions and wastes of the entire cycle. It can also identify abnormal emissions and energy consumptions and send warnings to management and regulators to make improvements to those processes.

[Insert Figure 1 here]

Built upon the life cycle component, the blockchain-based LCA system consists of two components: infrastructure and LCA. The infrastructure component serves as the technology foundation of the proposed system to collect, process, transmit, and store data. In this component, a variety of sensors are deployed at multiple phases of a product's lifecycle to collect ESG-related data in real time. However, using sensors to capture live emission data may not always be the most effective approach. For example, for industries like agriculture and stock farming, capturing real emissions requires considerable finances and time, and thus material balanced methods, a chain of custody approaches which can be used to trace the flow of materials through the value chain[8], have proved to estimate emissions more efficiently (Brentrup et al. 2000; Wang 2013).

Typical sensors related to LCA include electricity meters, water meters, RFID readers, heat meters, fiber optical sensors (Tao et al. 2014a), etc. Those sensors can be linked with the Internet or other communicating networks, which together compose an IoT infrastructure that can in a timely manner collect material flow and energy flow from the life cycle and directly upload to blockchains[9]. The distributed network of sensors and blockchain gather and integrate real-time data from a variety of procedures and locations in the life cycle, overcoming the data gathering difficulty faced by traditional LCA. Thanks to the append-only and immutable characteristics of blockchain, it can help prevent data manipulation by management and ensure data reliability.

Smart contracts play two roles in the system. First, they verify data collected from sensors upon uploading to the blockchain. Data authentication has been well recognized as a major concern for ESG reporting. For example, Toyota's subsidiary Hino recently admitted that it had falsified emission data by replacing purification equipment during emission evaluation tests.[10] Smart contracts combined with IoT have the potential to eliminate such a greenwashing practice. For example, each piece of equipment could be assigned an identification number, which would be registered and verified by smart contracts when uploading data to blockchain. If illicit replacement of equipment occured, smart contracts would detect the abnormal activity and prohibit the new equipment from uploading data onto the blockchain. Second, smart contracts control ESG reporting processes and enforce intervention once a problem is discovered. For example, if a company's reports are consistently delayed or materially misstated, smart contracts will be triggered to deduct its deposit or reduce its ESG credit rating, thus deterring companies from greenwashing and increasing incentives for companies to improve the quality of their ESG reports. Smart contracts can also quickly detect and stop fraudulent deeds. For example, if a company's greenhouse gas (GHG) emission exceeds a pre-set threshold in a smart contract, it will trigger a program to pause machines that are producing GHG or issue a fine to the company and report to ESG credit rating agencies.

To use the data on blockchain for LCA and other applications, it is necessary to extract and process the blockchain data, which is of heterogeneous data structures, to obtain valuable information. Robotic Process Automation (RPA)[11] can automatically extract data from blockchain and supplementary information from Enterprise Resource Planning (ERP) systems (such as product types when the data cannot be directly collected from sensors) and feed them into the LCA component. Then, the LCA component analyzes the input data (materials and energy consumption) together with preset parameters via LCA software to assess overall environmental impacts, such as emissions and wastes, from the entire product life cycle. Based on the output, the blockchain-based LCA system can further provide a variety of ESG services and applications, shown in the Services and Applications component in Figure 1.

One essential application of the system is cross-validation of ESG data between two companies by allowing the data to be successfully recorded on the blockchain only if the values entered by the two transaction parties match. For example, suppose company B consumes electricity provided by company A. The electricity meter at company A (company B) uploads the electricity sent to B (from A) onto the blockchain at the end of each day. A smart contract is set up to compare the electricity amounts uploaded from the two companies. If, due to the "first-mile" problem[12], company B underreports the electricity amount and thus fails to match with company A's electricity record, the smart contract will send a warning to the two companies. If a company is associated with many warnings, it will indicate there are many mismatching values between data uploaded by this company and those by its trading parties, and thus, further investigations should be performed to detect potential errors or fraud.

Another essential application is to generate an integrated ESG report to present the collective effect on the environment from all companies in a value chain. As shown in Figure 1, the value chain contains companies A, B, and C, and an integrated ESG report for the entire value chain is generated to reflect the collective environmental impact from those three companies. Such a collective effect can be derived from results produced by LCA software such as GREET[13]. For example, Table 1 shows a collection of emission metrics that are typically included in GREET results of what has been emitted during a product's life cycle and their volumes. Based on those data, an integrated ESG report can be generated by including metrics required by leading ESG reporting standards, such as GRI, NFRD, and SDG[14].

[Insert Table 2 here]

GRI Disclosures 305-1 to 305-7 require companies to report emissions of GHG regarding the metrics of CO2, CH4, N2O, HFCs, SF6, NF3, ODS, CFC-11, NOx, SOx, POP, VOC, HAP, and PM. NFRD recommends that companies report CO2 and GHG from four perspectives: direct GHG emissions; indirect GHG emissions from the generation of acquired and consumed electricity, steam, heat, or cooling; all other indirect GHG emissions that occur in the value chain of the reporting company; and GHG absolute emissions target. SDG mainly focuses on climate change (Goal 13), and its required metrics are related to GHG (see Table 1). Based on the metrics shown in Table 1, a potential integrated ESG report can be created by including a battery of emissions generated in different stages of a product life cycle that are required to be disclosed by those standards.

The last application is the validation of each company's ESG reports via tracing and identifying its corresponding LCA results (shown in Figure 1 using arrows pointing from the integrated report to individual companies for ESG report validation). Since blockchain records an entire flow history of materials and energy from one party to another, it can enable tracing of each material or energy transaction back to each company and product. Each transaction contains the detail of materials and energy a company purchased or consumed, based on which LCA results (such as GHG emissions) of a value chain can be apportioned to each company in the value chain. For example, if a company manufactures a product and its production volume is recorded on a blockchain, LCA can use the volume as an input and assess its environmental impacts, e.g., how much CO_2 this company emits to produce it. If the company has an upstream business partner who provides raw materials, LCA can assess CO_2 emissions from the partner by tracing material transactions on the blockchain regarding how many materials it sells to the manufacturing company and calculating CO_2 emissions in the extraction of those materials. If the company manufactures only one product, and the partner produces raw materials only for that company, the CO_2 emissions obtained from LCA can be used to validate the corresponding items in those companies' ESG reports. If a company produces several products, the CO_2 emissions of all its products from LCA should be aggregated and compared with what is reported in its ESG report. Following this method, an LCA-based ESG report can be generated from LCA output for each company in the value chain by aggregating the environmental impacts of all its products, which can be further used to validate the ESG report issued by the company. This process could be automated by RPA, which uses a set of robots to trace back to related records on the blockchain, extract energy and material data from the records, use them in LCA to assess the environmental impacts of each company, and compare the results with corresponding values in companies' ESG disclosures.

Finally, results of the ESG applications are uploaded to blockchain and reported to users. For example, organizations and government agencies that are concerned about global climate change can obtain a reliable summary of GHG emissions of the whole value chain and in a timely manner trace major emission sources, thus taking more proactive actions to reduce climate change. Different users (e.g., customers, investors, partners, auditors, NGOs, and governments) decide what, when, and to what extent the data should be reported according to their own needs. Consumers, for example, may demand complete information on a product's environmental impact and decide whether to purchase it. Investors may be interested in the environmental performance of a company to evaluate whether it has sustainable prospects to make investment decisions. Cooperative partners may assess if their supply chains and production processes meet environmental requirements. Auditors may need original and non-tampered data to perform future assurance tests. NGOs act as third parties to oversee all product life cycle stages and facilitate ESG-related trading markets such as carbon emission trading. Governments may monitor the environmental performance of enterprises in the entire value chain in order to develop better environmental protection policies.

4 Implementing Blockchain-based LCA System with Tesla's Electric Vehicles

The environmental impacts of Electric Vehicles (EV) can be quantified using LCA throughout their complete production and value chains (Hawkins, T.R. et al. 2013). We choose Tesla to illustrate the implementation of the blockchain-based LCA system because it is the biggest EV producer and provides sufficient disclosure in its public reports to be used as input data to LCA. Currently, no IoT data are available for LCA; therefore, we rely on Tesla's public data to simulate the data input process. In its 2020 Impact Report, Tesla disclosed the material and energy consumption and emissions in the manufacturing process of Model 3s (one type of EVs of Tesla) and their electricity consumption during use (Tesla Impact Report 2021). This section demonstrates the implementation and use of the blockchain-based LCA system to facilitate ESG disclosure regarding Model 3 across the entire value chain based on the data in Tesla's 2020 Impact Report.

We use the GREET software to perform LCA. GREET is a popular LCA software that simulates energy use, pollutant discharge, emission output, and water consumption of various vehicle and fuel combinations. GREET has two life cycle streams[15], as shown in Figure 2. One is the fuel cycle model (GREET1), which calculates energy use, pollutant discharge, emission output, and water consumption in the raw material extraction, transportation, refining, delivery, and use during vehicle operation. For Tesla Model 3, electricity is the main object of the fuel cycle. The raw material extraction process refers to the extraction of initial energy resources like oil, coal, or biomass to generate electricity. The transportation process refers to transporting energy resources to power stations. The refining process refers to the process of generating electricity. The delivery process is electricity transporting from power stations to charging facilities. The use process refers to the operation of Tesla Model 3. The other stream is the vehicle cycle model (GREET2), which explores a life cycle stream of energy use and emission output associated with the manufacturing and recycling of vehicles. The fuel and vehicle cycles together constitute the LCA model of EVs. Each process in the model requires input data such as the weight of coal and water, electricity consumption, etc.

[Insert Figure 2 here]

4.1 Smart Contracts

The first process of the blockchain-based LCA system is to post ESG-related information on a blockchain. A smart contract is created to automate several key functions in the process: 1) verification of ESG data sources, 2) monitoring ESG data (i.e., comparing data with pre-determined rules), and 3) reporting anomalies. We use the Ethereum[16] Ropsten Testnet[17] as the blockchain infrastructure and the Remix[18] online Integrated Development Environment (IDE) to create smart contracts. MetaMask[19] is used as a blockchain wallet, that is, software running on a blockchain that stores blockchain accounts and monitors all the transactions related to those accounts on the blockchain[20].

We define two rules and program them into smart contracts. The first rule is to verify the ESG data source. It is critical that only permissioned participants, e.g., companies and upstream and downstream partners in the value chain, should be allowed to upload data to the blockchain. Data from other sources are invalid, so they should be prohibited from being entered into the system. In the implementation, we set up two blockchain accounts, one of which (account address "0xb643d8DC8a77b18083C897095BE1e53E7C929250") is granted the permission of posting information on the blockchain, and the other (account address "0xb333d8DC8a77b18083C897095BE1e53E7C929222") does not have the permission. If data to be uploaded to the blockchain are from the permissioned account, they can be uploaded to the blockchain. Otherwise, a warning should be sent to the unauthorized account, and the data should not be posted on the blockchain.

The second rule is to monitor ESG data and report anomalies. Smart contracts can identify abnormal values based on pre-determined rules or standards; e.g., if a value sent to a smart contract is above a threshold, it will be considered abnormal, and a warning will be sent. Suppose the upper limit of normal electricity consumption to produce a certain auto part is 70 kWh. In this case, 70 kWh should be set as the threshold and coded into the smart contract. Then, if an electricity consumption of more than 70 kWh is sent to the smart contract, a warning "Electricity Value is Too High! Immediate actions should be taken" will be reported. This comparison function is also essential for the cross-validation application of the blockchain-based LCA system since nonconformity between relevant data reported from two companies regarding the same materials or products may indicate suspicious disclosure. For example, if company A records that it buys 100 kg coal from company B while company B records that it sells 200 kg coal to A, the nonconformity will indicate a potential error or fraud associated with this transaction, which should be prohibited from being recorded on-chain.

We create a smart contract with the two rules and deploy it on the Ethereum Ropsten Testnet network. Users of the system can see a data input form (shown in

Figure 3) that sends ESG-related information to the smart contract. They first enter their addresses to verify that they have permission to upload data to the blockchain. Figure 4 shows that a warning is reported when an unauthorized user tries to upload data to the system. For authorized users, they enter their addresses as well as electricity values into the form that posts the information on the blockchain. At this point, the smart contract compares the entered value with the pre-set threshold and shows a warning if the value exceeds the threshold (Figure 5 and Figure 6).

Note that in this implementation, electricity values are considered to be manually read from electricity meters and entered into the smart contract. The system could also link with IoT devices that automatically capture the data throughout the product life cycle and upload data to the blockchain.

[Insert Figure 3-6 here]

In addition, an overview of all records posted on the blockchain (Figure 7) and the detailed information about a specific transaction record (Figure 8) are automatically generated by the system. Each transaction record includes a block number, addresses of transaction parties, transaction value, time of posting, input (here, it is the electricity value, 65, highlighted in a red rectangle), etc. Thus, using this system, every interested party can see real ESG data collected reliably and in a timely manner from a variety of sources.

[Insert Figure 7-8 here]

4.2 LCA Using GREET

The second process of the blockchain-based LCA system is to obtain required data from the blockchain to feed into LCA software and assess the overall environmental impact of a product. Typical data to assess the environmental impact of EVs include raw materials (such as including steel, aluminum, copper plastics, glass and lithium, and cobalt for batteries), energy consumption, and emissions. Since we are unable to obtain all data in Tesla Model 3's life cycle, we use GREET and public information from Tesla's Impact Report to simulate the life cycle assessment. Content from the Tesla Impact report used in this study is summarized in Table 2 (Tesla Impact Report 2021).

[Insert Table 2 here]

In GREET, the vehicle cycle model (GREET2) contains the phases of raw material extraction, material processing, manufacturing and assembly, delivery, vehicle operation, and end of life. Following the common procedures in LCA analysis, we first calculate all energy use, pollutant discharge, emissions, and water consumption in the vehicle cycle (GREET2) before the vehicle operation phase[21] (by setting 0 km distance for the vehicle operation phase in the GREET software) (shown in Figure 9).

[Insert Figure 9 here]

The second step is to estimate electricity consumption during the fuel cycle. Tesla reports Model 3's emissions per mile when they are used in the U.S., Europe, and the Sichuan province of China. Those three regions use different power plants (such as coal, nuclear, oil, wind power, solar power, etc.) to generate electricity, which results in different types and volumes of emissions when driving EVs. Europe has the cleanest power plants of electricity, so the emissions per mile of Model 3 are the lowest in Europe. The function "Well-to-Pump" in GREET is used to analyze fuel production. For Model 3, this function analyzes the process from the raw material of electricity to delivery (EV charge stations) in the fuel cycle (GREET1).

Electricity in the United States generally comes from various types of power plants (natural gas, coal, oil, nuclear, etc.[22], each shown as a slice of the pie chart in

Figure 10), each of which has specific types of emissions. To estimate the emissions, we use default values provided by GREET that show the average emissions of a variety of gases produced during the production of electricity in the United States (Figure 11).

[Insert Figure 10 here]

[Insert Figure 11 here]

The last step of the Blockchain-based LCA System is to complete the vehicle cycle model (GREET2) by calculating electricity consumption, emissions output, and pollution discharge during the vehicle operation and end of life phases. The results are shown in Figure 12. Tesla Model 3 has no exhaust emission during driving because only electricity is consumed. Thus, all results of “Tailpipe emissions” in Figure 12 are zero.

[Insert Figure 12 here]

The final emission results are shown in Table 3. It contains air pollution metrics such as VOC, CO, and PM2.5 and GHG emissions such as CO_2 and CH_4. The column “WTP” represents emission output before the vehicle operation phase in the fuel cycle (GREET 1). The column “Vehicle Operation” shows the total emissions associated with vehicle operation only and does not include any upstream energy or emissions due to fuel manufacturing. Emissions produced in different phases like manufacturing and assembly and delivery are shown in columns containing “Components,” “ADR (Assembly Disposal and Recycling),” “Fluids,” and “Battery.” The column “Total” shows the combination of results from the vehicle and fuel cycles (overall LCA results). For example, according to Table 3, a Tesla Model 3 produces an average of 0.2625 kg/mi CO_2 in its life cycle.

[Insert Table 3 here]

5 Discussion, Limitations, and Future Research

Privacy could be a major concern of the blockchain-based LCA system when using it for ESG reporting. As a result, it is critical to decide whether information on the blockchain should be publicly or privately accessible. Blockchain technology can facilitate information sharing throughout the product life cycle; however, some stakeholders may be unwilling to share confidential information, such as their suppliers and material consumption, with their competitors or the general public, which may hinder system adoption. To alleviate this problem, the blockchain-based LCA should be implemented on a so-called "permissioned blockchain," which allows only authorized users to read and manage data on the blockchain. However, the fact that many stakeholders are involved in complex, dependent, or interdependent activities makes it challenging to design a reliable authorization regime. Future research could improve the proposed system by considering authorization processes based on the roles of different stakeholders.

Another possible limitation of the system is the accuracy of data captured by sensors before being posted to the blockchain. Inaccurate data could be captured due to technical limitations of sensors or deliberate intervention. For example, someone could spray water near air pollutant sensors to decrease PM2.5 concentration. Another example is the "diesel emissions scandal" in 2015, in which multiple car brands used a "defeat device" in their vehicles that hid emissions being emitted. Due to this deception, it was estimated that more than 500,000 cars imported into the UK were emitting 35 times more than the legal amount of noxious gases.[23] One possible solution is to conduct unscheduled spot checks for sensors. If a sensor is found to be dysfunctional, an alert will be posted to the blockchain that will block data transmitted from the sensor.

Finally, although smart contracts can automatically execute data processing and cross-validation, their innate disadvantage is that once deployed, they cannot be modified and changed. Thus, it might be costly for companies to adopt the system if their ESG reporting processes change frequently. Future research may explore solutions to this problem. One potential solution is to replace a large smart contract with a battery of small smart contracts. If a certain step in the process changes, only the small smart contract that is responsible for the changed step needs to be replaced by a new one rather than the entire large smart contract.

6 Conclusion

ESG reporting is currently facing several major problems such as greenwashing. Companies may greenwash their images; i.e., they falsely claim to make environmentally-friendly decisions through incomplete ESG disclosure such as stressing emissions only at the usage stage while ignoring emissions at the production and recycling stages, as in the beginning example. LCA offers a solution to this problem by examining the environmental impacts of a product from all relevant processes starting from inputs to outputs throughout the product's life cycle. However, a well-known challenge of LCA is to collect data in a reliable and effective manner. We introduce blockchain together with IoT to alleviate this concern by leveraging its advantages of decentralization, transparency, trustworthiness, and programmability.

This paper proposes a blockchain-based LCA system and illustrates its use to reliably collect and distribute ESG-related data and reflect the overall environmental impact from an entire value chain. We further implement the system by using a popular LCA software, the public information from Tesla's reports, and Ethereum. We demonstrate that the system can be useful for generating an integrated ESG report that presents environmental impacts from all companies in the EV value chain, cross-validates ESG information among participating companies, and traces back to the original information to validate the ESG report issued by each company.

This paper contributes to the ESG reporting literature by proposing a system that combines blockchain technology and LCA to achieve a synergy effect on facilitating ESG reporting. The paper also provides a discussion on the challenges in the adoption and use of the technologies, as well as potential solutions that could mitigate those concerns. Overall, our results shed light on utilizing emerging technologies to enable reliable, complete, and accurate ESG reporting.

Figures

Figure 3 Blockchain-based LCA System and Its Services and Applications

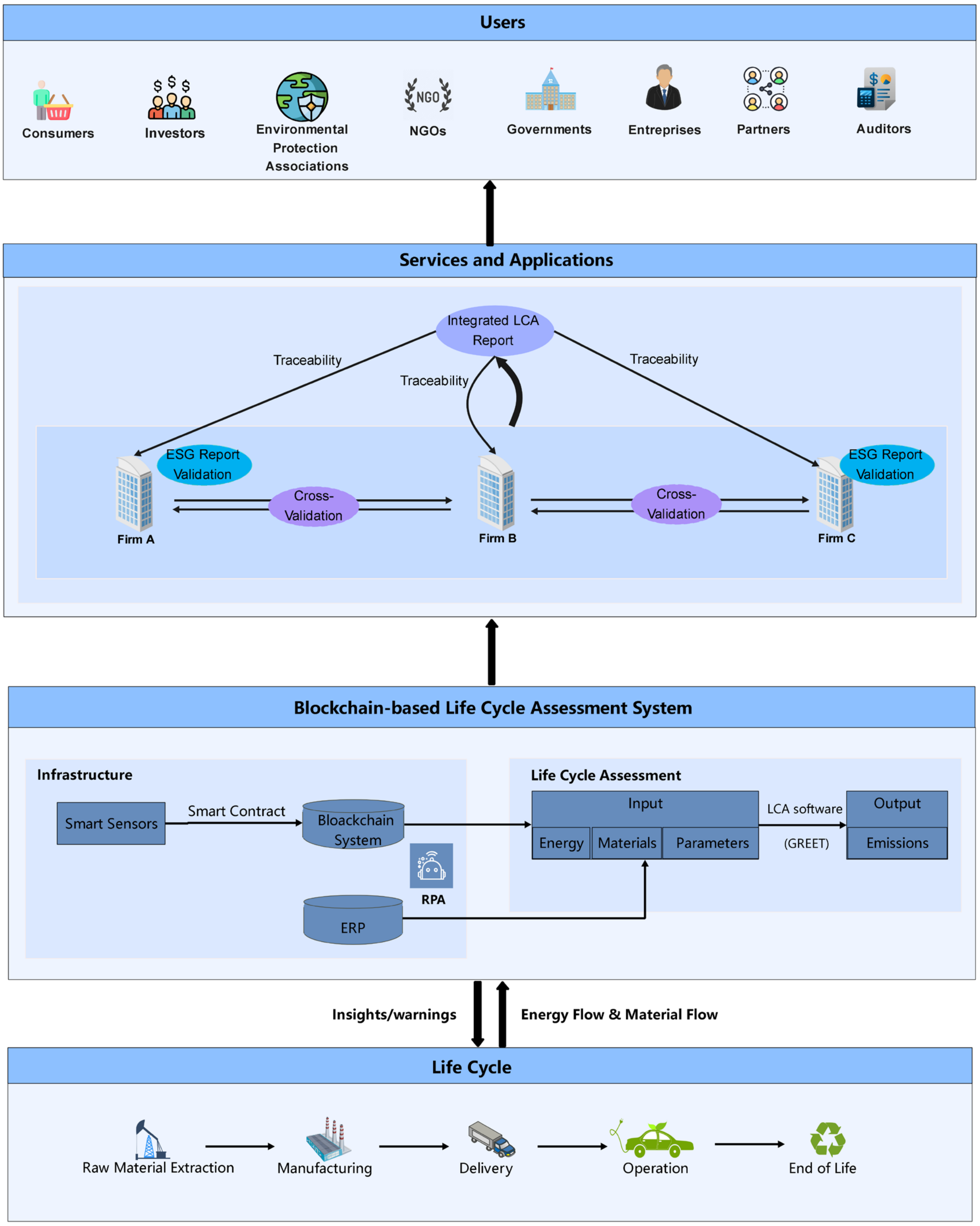


Figure 2 GREET Model

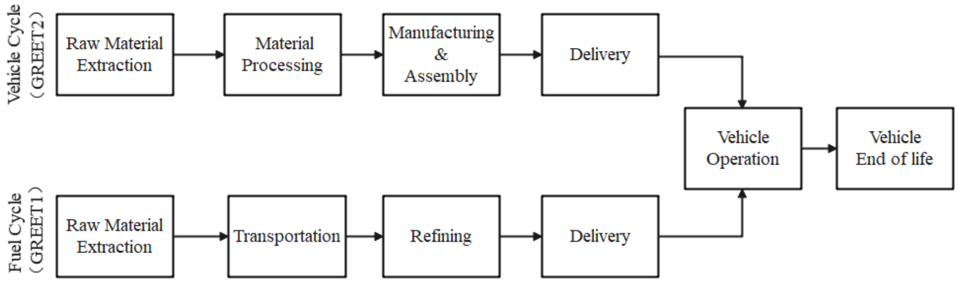


Figure 3 Data Input Window for Data Source Verification and Data Monitoring

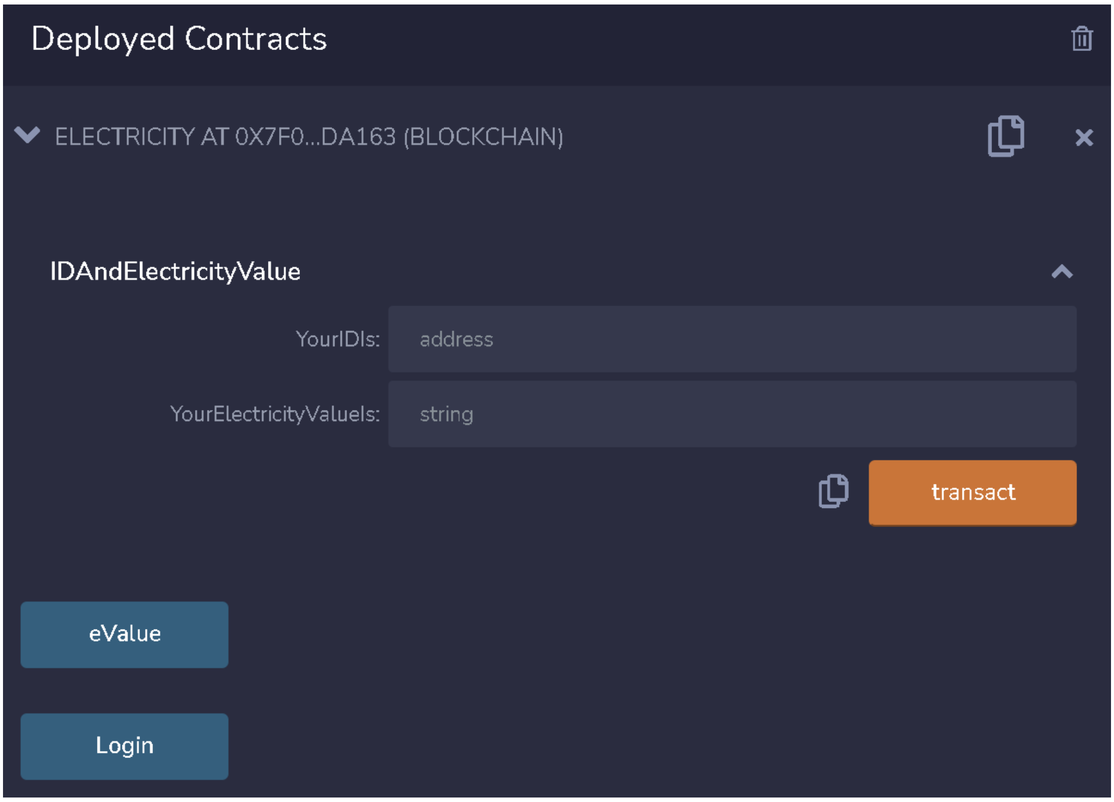


Figure 4 An Unauthorized Attempt Identified by Data Source Verification

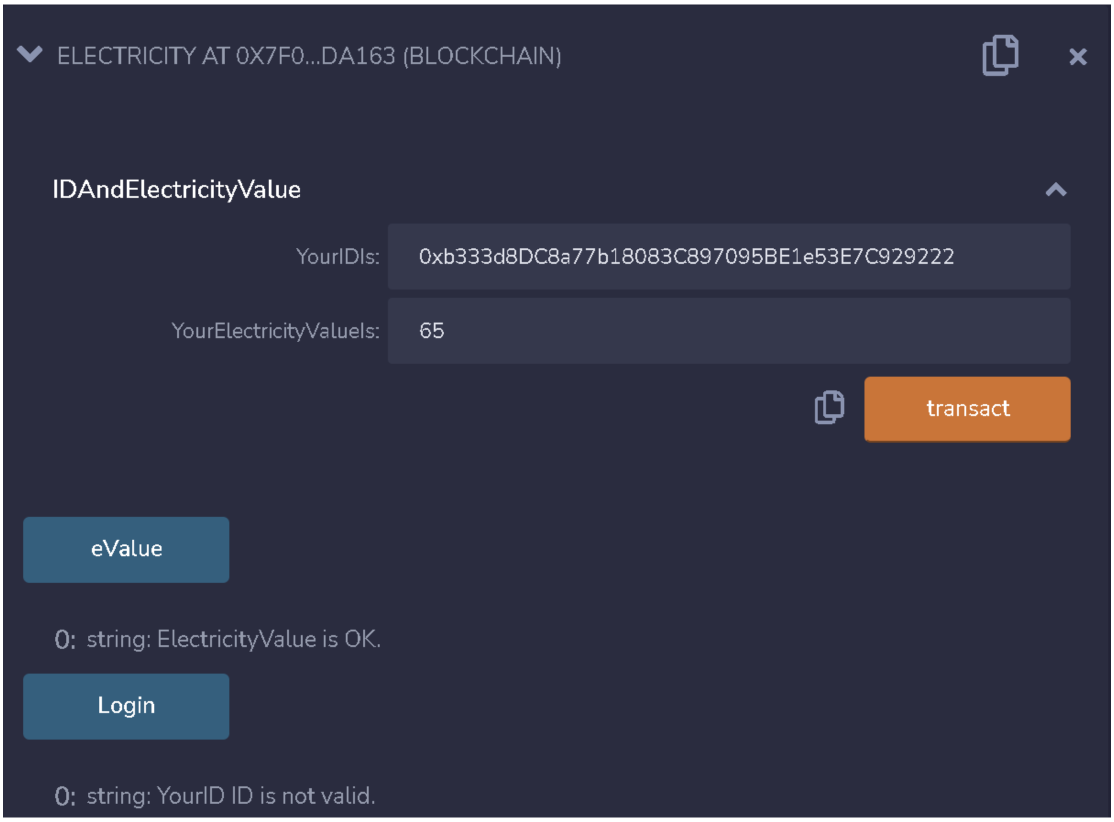


Figure 5 Data Monitoring and Anomaly Reporting

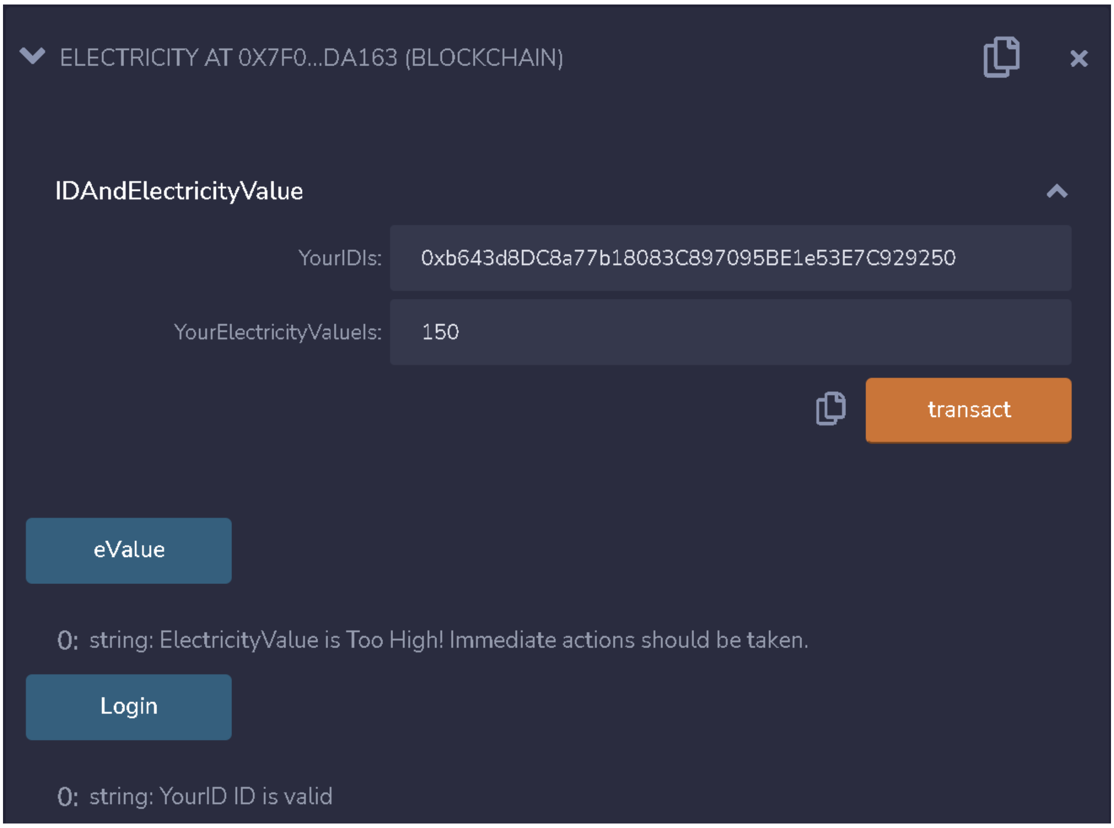


Figure 6 Data Monitoring in Normal Case

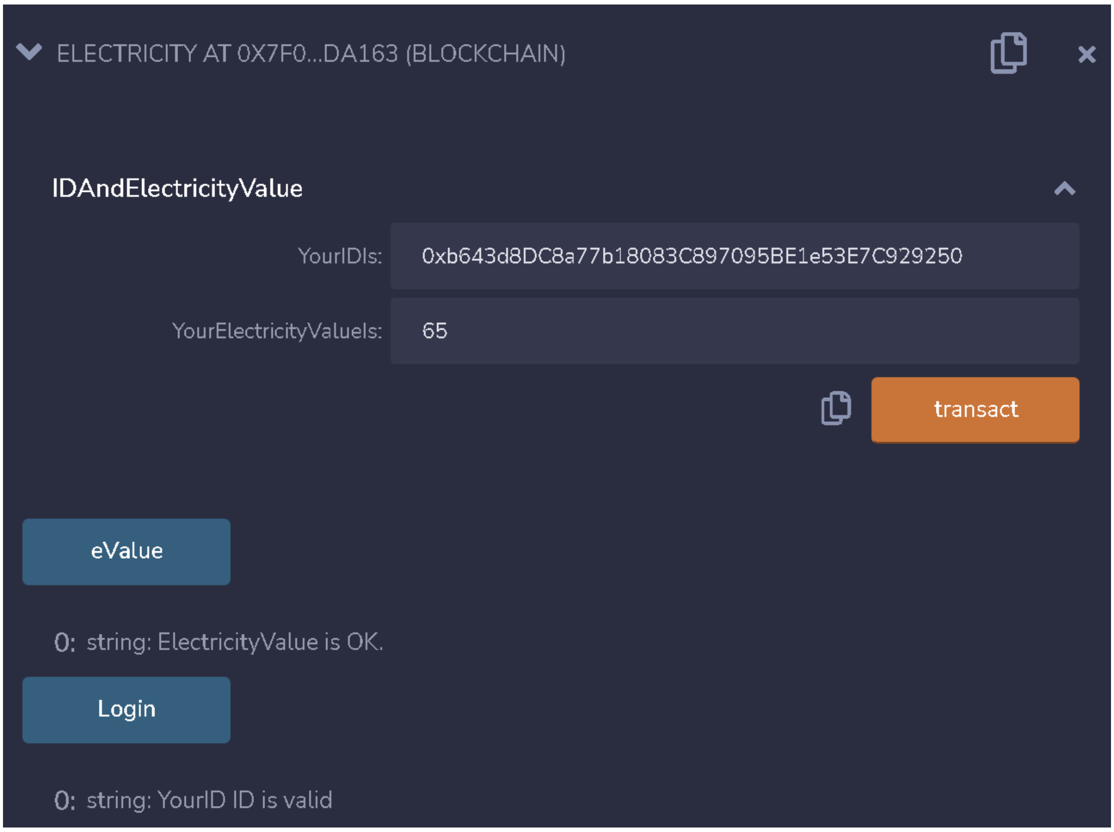


Figure 7 A Sample of Records on the Blockchain

Etherscan

Ropsten Testnet Network

All Filters | Search by Address / Txn Hash / Block / Token / Ens

Home | Blockchain | Tokens | Misc | Ropsten

Blocks

Network Utilization: 54.3% | Burnt Fees: 110,920.40 ETH

Block #12270199 to #12270223 (Total of 12,270,224 blocks)

First | < | Page 1 of 490809 | > | Last

Block	Age	Txn	Uncles	Miner	Gas Used	Gas Limit	Base Fee	Reward	Burnt Fees (ETH)
12270223	1 min ago	34	0	0x6784daf320b7ffb...	4,708,909 (58.86%, +18%)	8,000,000	3.02 Gwei	2.01399 Ether	0.014234 (50.42%)
12270222	3 mins ago	36	0	0x6784daf320b7ffb...	6,969,404 (87.12%, +74%)	8,000,000	2.76 Gwei	2.06168 Ether	0.019279 (23.81%)
12270221	3 mins ago	24	0	0x6784daf320b7ffb...	3,788,021 (47.35%, -5%)	8,000,000	2.78 Gwei	2.00574 Ether	0.010548 (64.76%)
12270220	4 mins ago	23	0	0xf19cb544ca1b912...	4,334,413 (54.18%, +8%)	8,000,000	2.75 Gwei	2.02905 Ether	0.011945 (29.13%)
12270219	4 mins ago	37	0	0x01679708348dd6...	7,746,955 (96.84%, +94%)	8,000,000	2.46 Gwei	2.01198 Ether	0.019112 (61.46%)
12270218	6 mins ago	18	0	0x6784daf320b7ffb...	7,990,723 (99.88%, +100%)	8,000,000	2.19 Gwei	2.06843 Ether	0.017527 (20.39%)
12270217	7 mins ago	24	0	0x4aec271789a155f...	7,980,978 (99.76%, +100%)	8,000,000	1.95 Gwei	2.03465 Ether	0.015569 (31.00%)
12270216	7 mins ago	74	0	0x6784daf320b7ffb...	1,862,426 (23.28%, -53%)	8,000,000	2.09 Gwei	2.00798 Ether	0.003893 (32.78%)
12270215	7 mins ago	5	0	0xf19cb544ca1b912...	173,194 (2.16%, -96%)	8,000,000	2.37 Gwei	2.00034 Ether	0.000411 (54.42%)

Figure 8 Detailed Information of a Record

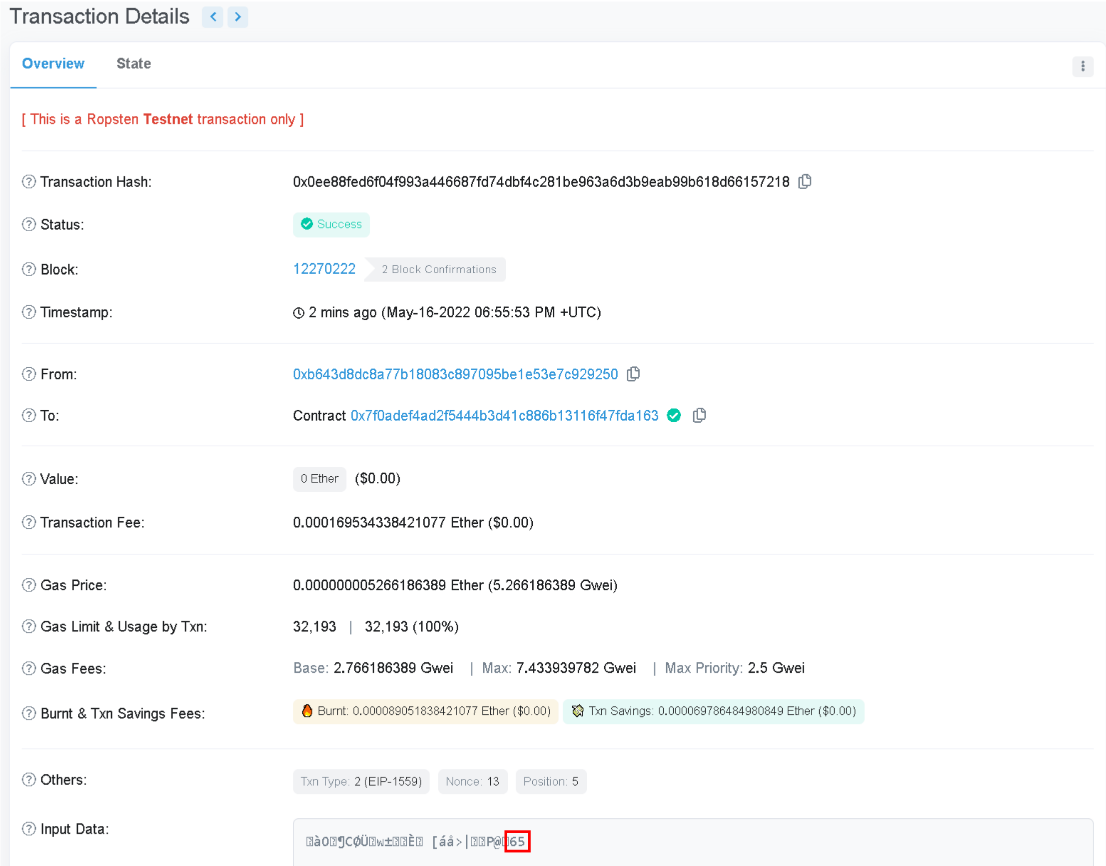


Figure 9 Consumption in Manufacturing Phase

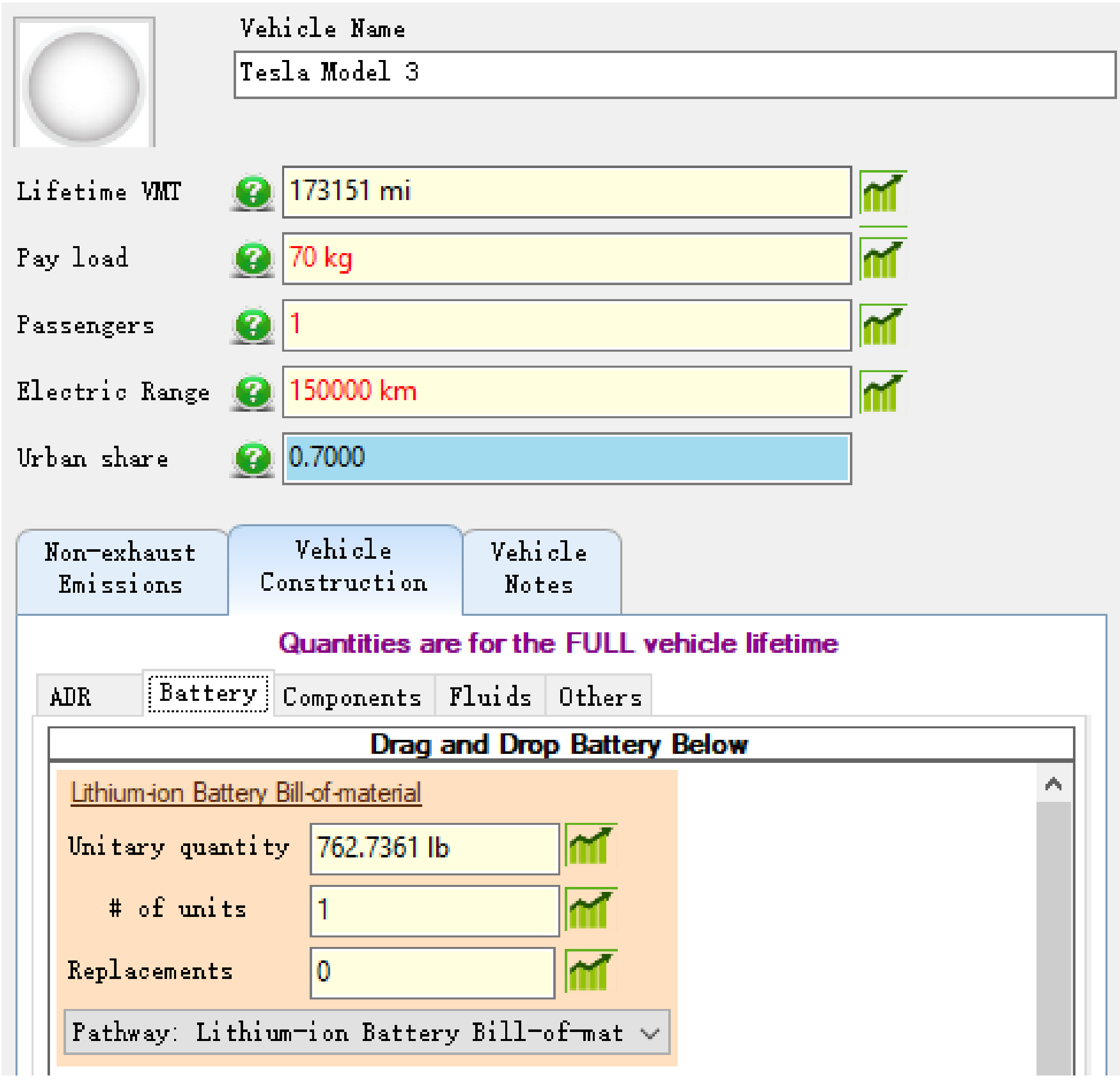


Figure 10 Distribution of Source of Electricity Production in the U.S. (Default Values from GREET)

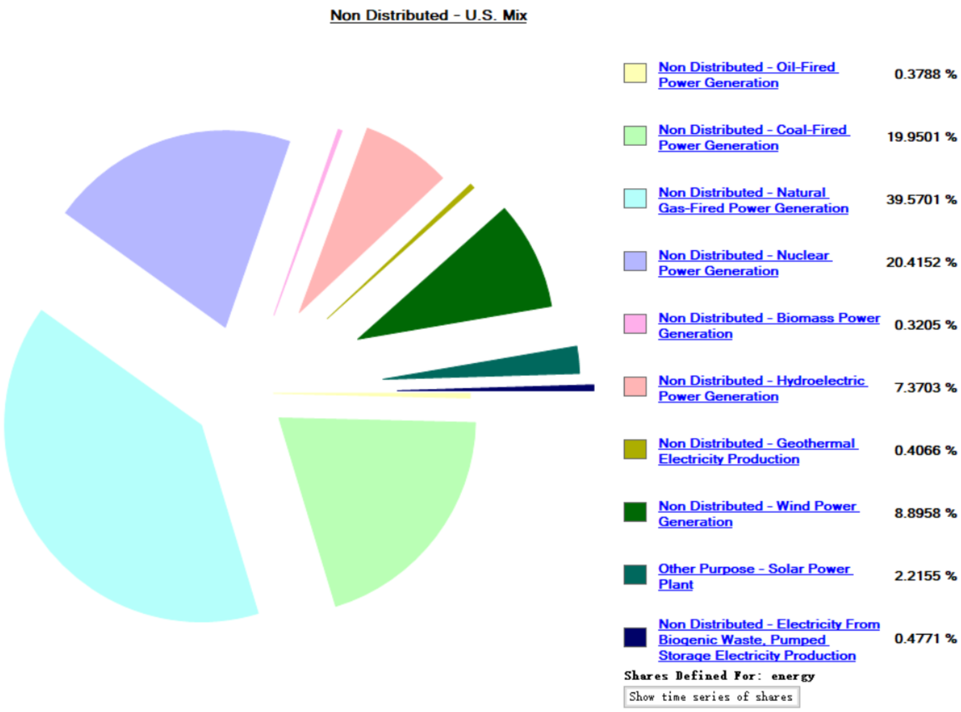
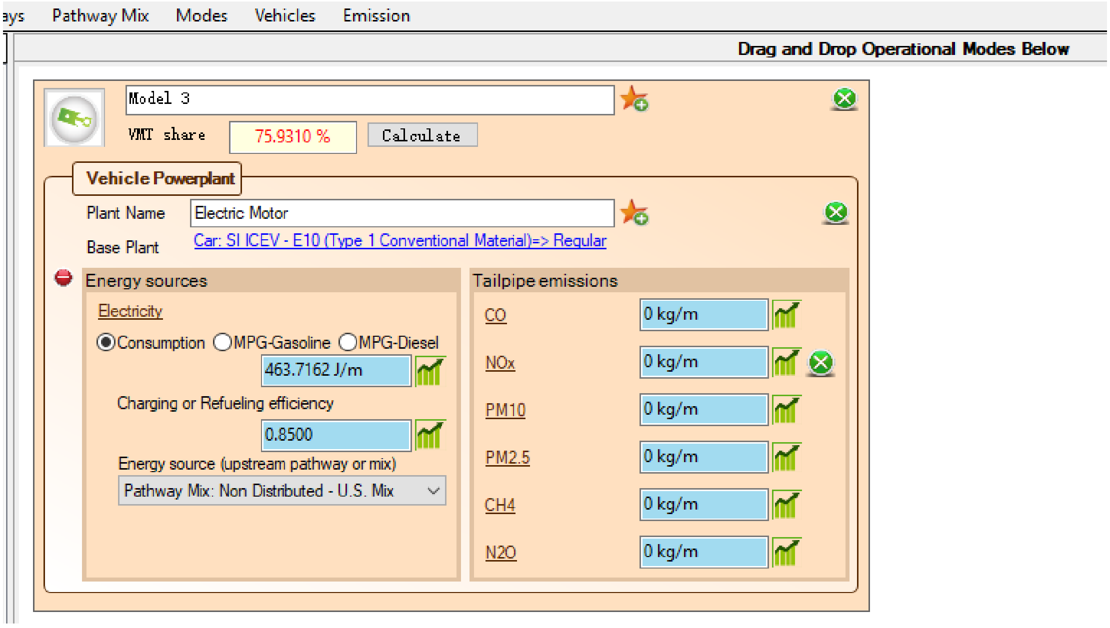


Figure 11 Various Gas Emission of Electricity Production

CO_2 Total	35.9378 g
CO_2	35.9555 g
CO_2_Biogenic	-1.7731e-5 kg
VOC	0.9073 mg
CO	9.9711 mg
NO_x	20.6663 mg
PM10	2.3611 mg
PM2.5	1.9815 mg
SO_x	20.3135 mg
CH_4	9.1298 mg
N_2O	0.6246 mg
BC	92.3389 ug
POC	0.5599 mg

Figure 12 Partial Results for Tesla Model 3



Tables

Table 1 Whether the Emission and Waste Metrics in GREET Report Are Required by ESG Guideline/Standards

Metrics in GREET report	Required by GRI	Required by NFRD	Required by SDG
Emission Metrics			
VOC (kg/mi)	√		
CH4_Biogenic (kg/mi)	√		
CO (kg/mi)			
NOx (kg/mi)	√		
PM10 (kg/mi)	√		
PM2.5 (kg/mi)	√		
SOx (kg/mi)	√		
CH4 (kg/mi)	√		
CO2 (kg/mi)	√	√	√
N2O (kg/mi)	√		
BC (kg/mi)			
POC (kg/mi)			
CO2_Biogenic (kg/mi)			
CO2_LandUseChange (kg/mi)			
GHG-100 (kg/mi)	√	√	√
VOC Urban (kg/mi)			
CH4_Biogenic Urban (kg/mi)			
CO Urban (kg/mi)			
NOx Urban (kg/mi)			
PM10 Urban (kg/mi)			
PM2.5 Urban (kg/mi)			
SOx Urban (kg/mi)			
CH4 Urban (kg/mi)			
CO2 Urban (kg/mi)			
N2O Urban (kg/mi)			
BC Urban (kg/mi)			
POC Urban (kg/mi)			
CO2_Biogenic Urban (kg/mi)			
CO2_LandUseChange Urban (kg/mi)			
BC_TBW (kg/mi)			
POC_TBW (kg/mi)			
PM10_TBW (kg/mi)	√		
PM2.5_TBW (kg/mi)	√		
VOC_evap (kg/mi)	√		
BC_TBW Urban (kg/mi)			
POC_TBW Urban (kg/mi)			
PM10_TBW Urban (kg/mi)			

PM2.5_TBW Urban (kg/mi)		
VOC_evap Urban (kg/mi)		
CF4 (kg/mi)		
C2F6 (kg/mi)		
CF4 Urban (kg/mi)		
C2F6 Urban (kg/mi)		
HFC-134a (kg/mi)	√	
SF6 (kg/mi)	√	
HFC-134a Urban (kg/mi)		
SF6 Urban (kg/mi)		
Waste Metrics		
Water Waste (m^3/mi)	√	√

Table 2 LCA Data Input for Tesla Model 3 (Derived from Tesla Impact Report [2021])

Manufacturing Phase Emissions and Wastes for Model 3	
Activities Examined	**Activities Not Examined**
Raw and semi-finished material production	Capital goods (e.g., machinery, buildings), infrastructure (e.g., roads, power transmission systems),
Transportation	Employee commute
Mechanical processing and shaping	External charging
Battery manufacturing	Equipment and infrastructure
Vehicle assembly and paint shop	Maintenance and service during use
All fuels and energy (electricity, natural gas, etc.)	Packaging
Other auxiliaries (lubricants, water, etc.)	Transport to recycler
End-of-life disposal	Disposal of manufacturing waste, inbound transportation from Tier 1 suppliers, and distribution to customers

Use Phase Emissions and Wastes for Model 3

- Use phase emissions for grid charging are based on Model 3 delivery-weighted state-level grid mix based on DOE（the U.S. Department of Energy's） estimates of state-level grid carbon intensity (https://afdc.energy.gov/vehicles/electric_emissions.html)
- Emissions of ~120 gCO2e/mi are a result of calculating the geographic distribution of the Model 3 in the U.S. based on Tesla's delivery data which weights state-level carbon intensity figures and assumes no change in grid mix into the future.
- The vast majority of generated waste such as paper, plastics, and metals, is recyclable. At Gigafactory Shanghai, for example, just 7% of total waste generated in 2021 was not recyclable

Table 3 A Potential Integrated LCA Report for Tesla Model 3

Integrated LCA report							
Product	Tesla Model 3						
Period	2022.01.01-2022.01.21						
Metrics	**WTP**	**Operation Only**	**Total**	**Components**	**ADR-Vehicle (Assembly, Disposal, and Recycling)**	**Fluids**	**Battery**
Total Energy (J/mi)	697587.26	2588750	3838739	367583.1	86295.1117	66272.84	32251.14
CH4 (kg/mi)	0.0002811	3.74E-06	0.000369	5.86E-05	0.000012901	7.38E-06	5.15E-06
CO2 (kg/mi)	0.0448	0.1858	0.2625	0.022	0.0049	0.0033	0.0017
VOC (kg/mi)	7.205E-05	2.15E-05	0.000283	2.02E-05	9.9673E-06	0.000159	6.18E-07
SOx (kg/mi)	1.757E-05	1.12E-06	0.000284	0.000119	2.1494E-06	8.16E-06	0.000136
N2O (kg/mi)	7.077E-06	3.86E-06	1.17E-05	5.33E-07	1.2289E-07	7.89E-08	6.27E-08
NOx (kg/mi)	6.213E-05	3.3E-05	0.000131	2.48E-05	4.4854E-06	4.34E-06	2.09E-06
CO (kg/mi)	3.829E-05	0.0015	0.0016	0.000106	2.9411E-06	1.65E-06	2.99E-06
PM10 (kg/mi)	6.984E-06	4.49E-06	2.65E-05	1.16E-05	7.7885E-07	7.11E-07	1.92E-06
PM2.5 (kg/mi)	4.686E-06	3.97E-06	1.58E-05	5.43E-06	4.7067E-07	3.77E-07	8.34E-07
BC (kg/mi)	6.258E-07	2.8E-06	3.64E-06	1.38E-07	3.09E-08	2.98E-08	1.41E-08
POC (kg/mi)	1.009E-06	9.93E-07	2.47E-06	3.03E-07	1.0079E-07	3.44E-08	3.18E-08
Water Waste (m^3/mi)	0.0004889	0	0.000646	0.000125	1.66874E-05	5.13E-06	1.08E-05

[1] https://fortune.com/2021/11/22/esg-investing-needs-standards-to-prevent-fraud-and-greenwashing/
[2] Tesla Removed from S&P 500 ESG Index on Lack of Low Carbon Strategy, Controversy Risk - ESG Today
[3] Source: https://www.zemo.org.uk/search.htm
[4] Source: https://www.theguardian.com/environment/2021/aug/20/electric-car-batteries-what-happens-to-them
[5] European Financial Reporting Advisory Group. (2021). Current non-financial reporting formats and practices. European financial reporting advisory group website. https://www.efrag.org/Assets/Download?assetUrl=%2Fsites%2Fwebpublishing%2FSiteAssets%2FEFRAG%2520PTF-NFRS_A6_FINAL.pdf
[6] https://ethereum.org/en/
[7] https://docs.soliditylang.org/en/v0.8.13/
[8] https://plasticseurope.org/knowledge-hub/mass-balance-approach-to-accelerate-the-use-of-renewable-feedstocks-in-chemical-processes/
[9] Blockchain here can be considered as a kind of distributed database that can avoid the conflicts when multiple modifications are made simultaneously by different computers in the distributed database system.
[10] Source: https://www.abc.net.au/news/2022-03-08/hino-admits-cheating-emissions-fuel-economy-tests/100890366?utm_campaign=abc_news_web&utm_content=link&utm_medium=content_shared&utm_source=abc_news_web
[11] RPA is a form of business process automation technology based on metaphorical software robots operating on the user interface of other computer systems in the way a human would do (Van der Aalst et al. 2018).
[12] Alles & Gray (2020) defines "the first mile problem" as the distinction between the underlying transactions recorded on the blockchain database and the actual physical events.
[13] Source: https://greet.es.anl.gov/
[14] According to European Financial Reporting Advisory Group (2021), GRI, NFRD, and SDG are the three most widely used ESG reporting standards.
[15] https://greet.es.anl.gov/
[16] https://ethereum.org/en/
[17] Source: https://ropsten.etherscan.io/
[18] Source: https://explorer.testnet.rsk.co/
[19] Source: https://metamask.io/
[20] https://www.softwaretestinghelp.com/what-is-a-blockchain-wallet/#:~:text=The%20blockchain%20ID%20is%20the,digital%20assets%20on%20the%20blockchain.
[21] The vehicle operation phase is analyzed separately.
[22] The types of power plants are provided by GREET.
[23] Source: https://www.bbc.com/news/business-34324772

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